TCGA-BRCA Dataset

WGCNA, Module Preservation, Hub-Gene Identification

Report (Part 01)

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Dataset: TCGA BRCA Gene Expression
Network Configurations: Signed and Unsigned
Tools: R, WGCNA, DESeq2, clusterProfiler, Cytoscape

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0 Contents

1 Differential Expression Analysis

1.1 Motivation

Before constructing co-expression networks, it is necessary to identify which genes are actually behaving differently between tumour and normal breast tissue. Running WGCNA on all 19,938 protein-coding genes indiscriminately would include thousands of genes whose expression is essentially flat across all samples, contributing only noise to the correlation structure. Our preferred pipeline therefore uses differential expression analysis as a biologically motivated gene selection step: only genes that are statistically confirmed to change between conditions are carried forward into network construction.

1.2 What Differential Expression Analysis Is

Differential expression (DE) analysis asks, for each gene independently, whether the observed difference in mean expression between two groups (here, Primary Tumour versus Solid Tissue Normal) is large enough and consistent enough across samples to be considered real rather than a product of random sampling variation.

DESeq2 was used for this analysis. It models raw integer read counts using a **negative binomial distribution**. RNA-seq data consists of counts: the number of times each gene's RNA molecule was read by the sequencer. Counts are whole numbers, never negative, and their statistical behaviour differs from the continuous measurements that simpler methods assume. A natural first choice for count data is the Poisson distribution, which has one convenient property: its variance equals its mean. However, RNA-seq data consistently violates this. **A gene with mean expression 100 might have variance of 500 or 2,000 across samples**, far exceeding what pure counting noise would predict. This excess variability arises because biological samples are never identical even within the same condition: tumour samples differ in genetic background, tumour microenvironment, and cell type composition. This phenomenon is called **overdispersion**.
The negative binomial distribution extends the Poisson with an extra parameter that allows variance to exceed the mean, making it the correct statistical model for overdispersed RNA-seq count data. For each gene, DESeq2 fits a generalised linear model using this distribution and applies a **Wald test** to assess whether the estimated fold change between conditions is significantly different from zero.
The Wald test produces a signal-to-noise statistic by dividing the estimated fold change by its standard error (a measure of how uncertain the estimate is). If a gene has a large fold change and that estimate is precise (small uncertainty), the Wald statistic is large, producing a small p-value. If the fold change is large but the estimate is uncertain (because the gene has very few counts), the Wald statistic is small and the result is not called significant. This is what makes DESeq2 more reliable than simply comparing averages: **it accounts for how much evidence supports each estimate**.

1.3 Combined Matrix and Experimental Design

The tumour and normal raw count matrices were column-bound into a single combined matrix of 19,938 genes $\times$ 1,178 samples. A metadata table was constructed assigning each sample a `Condition` label of either `Tumor` or `Normal`, with `Normal` set as the reference level. The design formula `design = ~Condition` instructs DESeq2 to model gene expression as a function of tissue condition, so that all fold change and significance estimates are computed relative to the normal baseline.

1.4 Statistical Measures

1.4.1 Size Factor Normalisation (Internal to DESeq2)

Before any statistical testing, DESeq2 corrects for differences in sequencing depth across samples using the **median-of-ratios method**. **Different samples are sequenced to different depths**: one sample might yield 20 million total reads while another yields 80 million. To correct for this, DESeq2 computes a **size factor** for each sample. For every gene, it calculates the geometric mean count across all samples (a type of average that works well for ratios). It then computes the ratio of each sample's count to that geometric mean, for every gene. The median of all these ratios for a given sample is that sample's size factor. The median is used rather than the mean because some genes will be genuinely differentially expressed, making their ratios biologically inflated or deflated. The median ignores these outliers and captures the typical ratio, which reflects sequencing depth rather than biology. All counts are divided by their sample's size factor before model fitting. This step is performed entirely internally by `DESeq()` and ***does not require any manual pre-processing***; raw integer counts are the correct and required input.

1.4.2 $\log_2$ Fold Change

The primary effect size measure is the $\log_2$ fold change ($\log_2$FC), defined as:

$$\log_2 \mathrm{FC} = \log_2\left(\frac{\text{mean normalised expression in Tumour}}{\text{mean normalised expression in Normal}}\right)$$

A value of $+1$ indicates a **doubling** of expression in tumour relative to normal; -1 indicates expression is **halved**; $+2$ indicates a fourfold increase. The $\log_2$ scale is used rather than a raw ratio because it makes upward and downward changes symmetric around zero. On a raw ratio scale, a gene that doubles gives 2.0 while a gene that halves gives 0.5 – these are not symmetric around 1. On the $\log_2$ scale, doubling gives $+1$ and halving gives -1, which are symmetric around zero and far easier to interpret and visualise. DESeq2 applies **empirical Bayes shrinkage** to $\log_2$FC estimates, which means that for genes with very low counts or high variability, the fold change estimate is pulled toward zero. This is because low-count genes produce unreliable fold change estimates: a gene with counts of 1 and 3 looks like a threefold change, but this could easily be noise. Shrinkage reduces these inflated estimates, making the results more statistically trustworthy even if some fold changes appear smaller than a naive calculation would suggest.

1.4.3 Wald Test and Raw *p*-value

For each gene, DESeq2 computes a Wald statistic by dividing the shrinkage-adjusted $\log_2$FC by its standard error:

$$W = \frac{\log_2 \mathrm{FC}}{\mathrm{SE}(\log_2 \mathrm{FC})}$$

The standard error measures how uncertain the fold change estimate is. A gene with thousands of counts across all samples has a very precise estimate (small SE), while a gene with sparse counts has a highly uncertain estimate (large SE). The Wald statistic is therefore a signal-to-noise ratio: a large fold change that is precisely estimated gives a large W, while a large fold change with high uncertainty gives a small W. Under the null hypothesis that the true fold change is zero, W follows a standard normal distribution (a bell curve centred at zero with standard deviation one). The raw p-value is the probability of observing a value of $|W|$ at least as large as the one computed, purely by chance, if there were truly no difference between conditions. A small p-value means the observed fold change is unlikely to be random noise.

1.4.4 Multiple Testing Correction and Adjusted p-value ($padj$)

Because DE testing is performed simultaneously for all 19,938 genes, using raw p-values directly would produce a very large number of false positives. At a threshold of $p < 0.05$, approximately $19{,}938 \times 0.05 \approx 997$ genes would be called significant by chance alone even if no gene were truly differentially expressed. This is the **multiple testing problem**: the more tests you run simultaneously, the more false positives accumulate even at stringent thresholds.

DESeq2 addresses this using the **Benjamini-Hochberg procedure**, which controls the **false discovery rate (FDR)**: the expected proportion of false positives among all genes declared significant. The adjusted p-value ($padj$) is the quantity output by this procedure. A threshold of $padj < 0.05$ means that among all genes called significant, at most 5% are expected to be false discoveries, regardless of how many total tests were conducted. If 4,000 genes pass this threshold, at most 200 of them are expected to be false positivesa, which is a controlled and defensible standard. Genes for which DESeq2 could not compute a $padj$ (due to zero counts across all samples, etc) are assigned `NA` and are removed from the results table before filtering.

1.4.5 Filtering Criteria

Two thresholds were applied jointly to define the final set of differentially expressed genes (DEGs):

1. $padj < 0.05$: statistical significance after multiple testing correction, ensuring the false discovery rate is controlled at 5%.
2. $|\log_2 \mathrm{FC}| \geq 1$: biological significance, requiring at least a twofold change in expression in either direction. This threshold is necessary because with over 1,000 samples, DESeq2 has sufficient statistical power to detect even trivially small expression differences as highly significant. A gene that changes by only 10% between tumour and normal may achieve $padj < 0.001$ at this sample size while carrying no meaningful biological signal. The $|\log_2 \mathrm{FC}| \geq 1$ criterion ensures that only genes with a substantial magnitude of change are retained.

Applying both thresholds jointly yielded **5,157 significant DEGs**. The full DE results table was saved to `TCGA-BRCA_DE_results_tumor_vs_normal.csv` and the list of significant DEG Ensembl IDs to `TCGA-BRCA_sig_DEGs_list.csv`.

2 DESeq2 Normalisation and Variance-Stabilising Transformation

2.1 Why a Second Normalisation Step Is Needed

Once the DEG set has been identified, the data must be transformed into a form suitable for WGCNA. **WGCNA operates by computing pairwise Pearson correlations between gene expression profiles across samples**. Raw counts are not appropriate for this because they are on a highly skewed scale: a gene with a mean count of 10,000 has inherently larger absolute fluctuations than a gene with a mean count of 50, even if the proportional variability is identical. This means correlation values computed on raw counts are dominated by highly expressed genes and are distorted by count-dependent variance. Two transformation steps are therefore applied: size factor normalisation followed by variance-stabilising transformation.

This second DESeq2 run is entirely separate in purpose from the DE analysis. The design formula is set to `design = ~1`, meaning no group comparison is performed. DESeq2 is used purely to estimate size factors for depth correction and to produce a fitted dispersion model, which is required as input to the variance-stabilising transformation function. Crucially, the tumour and normal matrices are processed independently at this stage, because WGCNA will construct separate co-expression networks for each tissue type and their expression scales should not be conflated.

2.2 Size Factor Normalisation

The `DESeq()` function with `design = ~1` estimates one size factor per sample using the median-of-ratios procedure described in Section 3.4.1. Normalised counts are extracted via `counts(dds, normalized = TRUE)`, yielding a matrix where count differences due to sequencing depth have been removed and expression values are comparable across samples within each tissue type.

2.3 Variance-Stabilising Transformation

Even after depth correction, normalised counts remain on a scale where variance is not independent of the mean: genes with high expression tend to show higher absolute variance than lowly expressed genes. The consequence for WGCNA is that a highly expressed gene, simply by virtue of spanning a wider numerical range, will produce stronger correlations with other genes than a lowly expressed gene showing the same proportional co-variation. The correlation structure would therefore be distorted toward high-expression genes.

The **variance-stabilising transformation (VST)** corrects for this. Using the mean-variance relationship estimated by DESeq2 across all genes, it applies a gene-specific mathematical transformation that compresses high-expression genes and expands low-expression genes until all genes have approximately equal variance (**homoskedastic distribution**) regardless of their expression level. The output values are on an approximately log-like continuous scale (will see values like 8.3 or 11.7 rather than whole number counts).

2.4 Gene Set Used for WGCNA

In the standard Nguyen and Zeng (2025) protocol, a secondary variance filter retaining the top 5% most variable genes (95th percentile of row variance) is applied after VST to further reduce the gene set before network construction. However, this protocol begins from all protein-coding genes (∼19,000), so the top 5% represents approximately 950 genes, a reasonable WGCNA input.

In the present pipeline, the 5,157 DEGs were already a pre-filtered biologically meaningful gene set before VST was applied. Applying the 95th percentile variance filter on top of this pre-filtered set would retain only 5% of 5,157 $\approx$ 258 genes per tissue, and the intersection of the two tissue-specific filtered sets collapsed further to just 94 genes, which is **far too small for reliable co-expression network construction**. Applying a stringent variance filter on top imposes a redundant selection that dramatically over-reduces the gene set.

The secondary variance filter was therefore omitted. All 5,157 VST-transformed DEGs were carried forward directly into WGCNA. This is methodologically appropriate because the DEG criterion already ensures that every retained gene shows a statistically significant and biologically substantial expression change between conditions; additional variance filtering provides no further benefit and causes substantial information loss.

2.5 Gene Alignment Between Tumour and Normal

The VST is applied independently to the tumour and normal DEG matrices. Since both matrices were subset to the identical set of 5,157 DEG Ensembl IDs before VST, the row sets are already matched. An explicit intersection step was nevertheless retained as a safety check. Both matrices were confirmed to share the same 5,157 genes and to contain no missing values before proceeding to network construction. The final matrices passed into WGCNA are therefore:

Matrix	**Genes**	**Samples**
`filtered_tumor`	5,157	1,065
`filtered_normal`	5,157	113

3 Soft-Thresholding Power Selection

3.1 Scale-Free Topology and the Need for a Soft Threshold

Scale-free networks in biology: Real biological networks, including gene regulatory networks, follow a scale-free topology, meaning that the distribution of node connectivity (number of co-expression partners per gene) follows a power law: $P(k) \sim k^{-\gamma}$. A small number of "hub" genes have very many connections, while the majority of genes have few.

The soft-thresholding approach: WGCNA does not build hard binary networks (gene A either is or is not connected to gene B). Instead, it raises the Pearson correlation between every gene pair to an exponent β to compute a continuous adjacency value. For a signed network:

$$A_{ij} = \left(\frac{1 + r_{ij}}{2}\right)^{\beta}$$

For an unsigned network:

$$A_{ij} = |r_{ij}|^{\beta}$$

Raising to β exponentially penalises weak correlations (driving them toward zero) while preserving strong correlations.

The function `pickSoftThreshold()` evaluates a range of candidate powers (1-20 generally) and reports the Scale-Free Topology Model Fit (R^2) and mean connectivity for each. We select the *lowest* power at which $R^2 \geq 0.90(/0.85)$ and the slope of the connectivity distribution is negative, ensuring that the network behaves like a true biological system.

3.2 Signed vs. Unsigned Networks: A Critical Distinction

As mentioned above, an unsigned nmetwork uses $|cor(i, j)|^{\beta}$ i.e., both positive and negative correlations contribute equally to connectivity. A signed network maps this correlation from $[-1, 1]$ to $[0, 1]$ before raising to the power β, so the negatively correlated genes get near-zero weight.
Signed networks therefore typically require higher β than unsigned because the signed transformation compresses the range of the correlation values more gently, with lesser contrast betweeen weak and strong positive correlations. A larger exponent is needed to achieve the same degree of differentiation between strong and weak connections.

3.3 Selected Candidate Powers

After running `pickSoftThreshold()` separately for each combination of tissue type and network type, the following candidate powers were validated:

Configuration	**Power (β)**
Tumor Unsigned	**4**
Tumor Signed	**9**
Normal Unsigned	**7**,8
Normal Signed	**14**,16

3.3.1 Why multiple 'candidate' powers?

Sometimes, multiple powers in the sft output seem to fit in within our **desired range of optimums**, so it is best to evaluate all the networks for these powers to choose the best one for our dataset.

3.3.2 Difference in values of powers

The substantially higher powers required for the normal dataset compared to the tumour dataset reflect a fundamental biological difference: cancer cells are driven by aggressive oncogenic signals that force hundreds of genes into very high mutual correlation, making scale-free topology easy to achieve. Normal tissue operates under tightly regulated homeostatic control, producing more subtle and distributed correlations that require a stronger mathematical penalty to reveal the underlying network structure.

4 Figure 1: Scale Independence and Mean Connectivity Plots

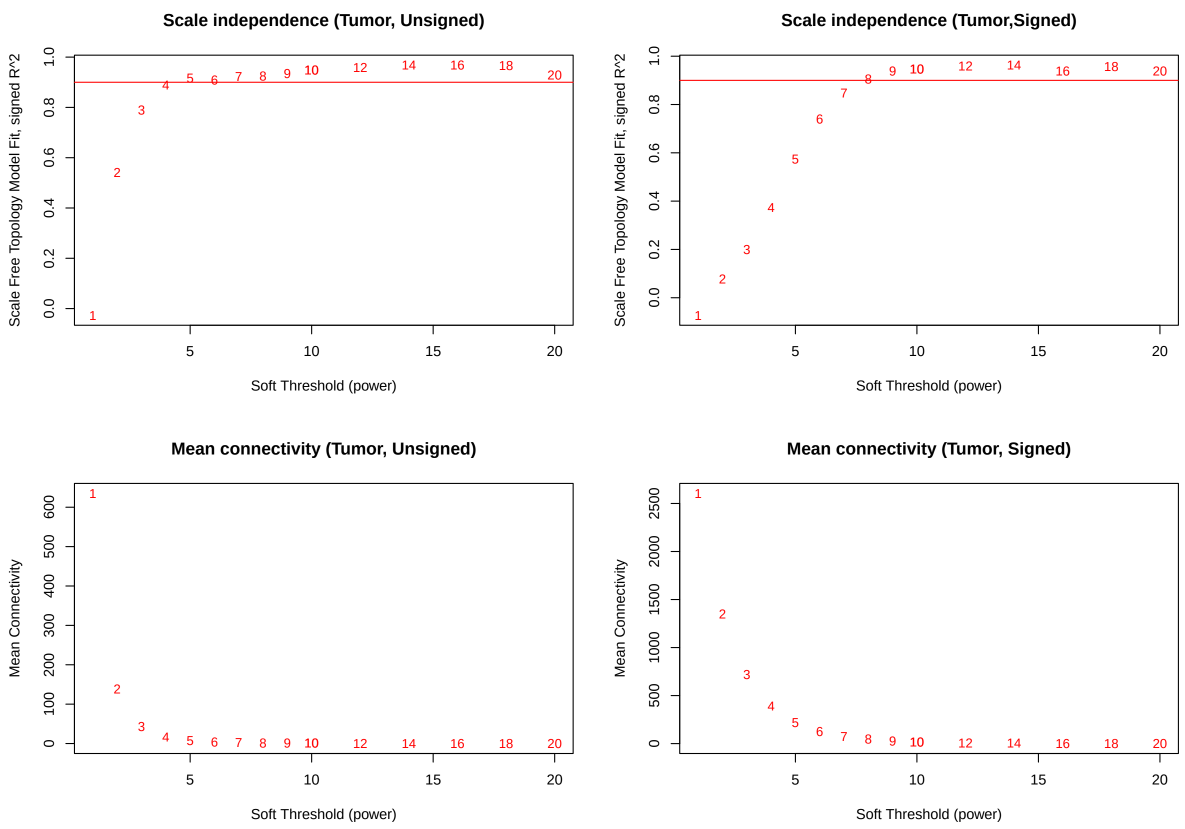


Figure 1: Scale independence (signed R^2) and mean connectivity plots for the **Tumor** dataset, showing both unsigned (left two panels) and signed (right two panels) configurations. Power numbers are plotted as red text labels. The horizontal dashed line indicates the $R^2 = 0.90$ threshold.

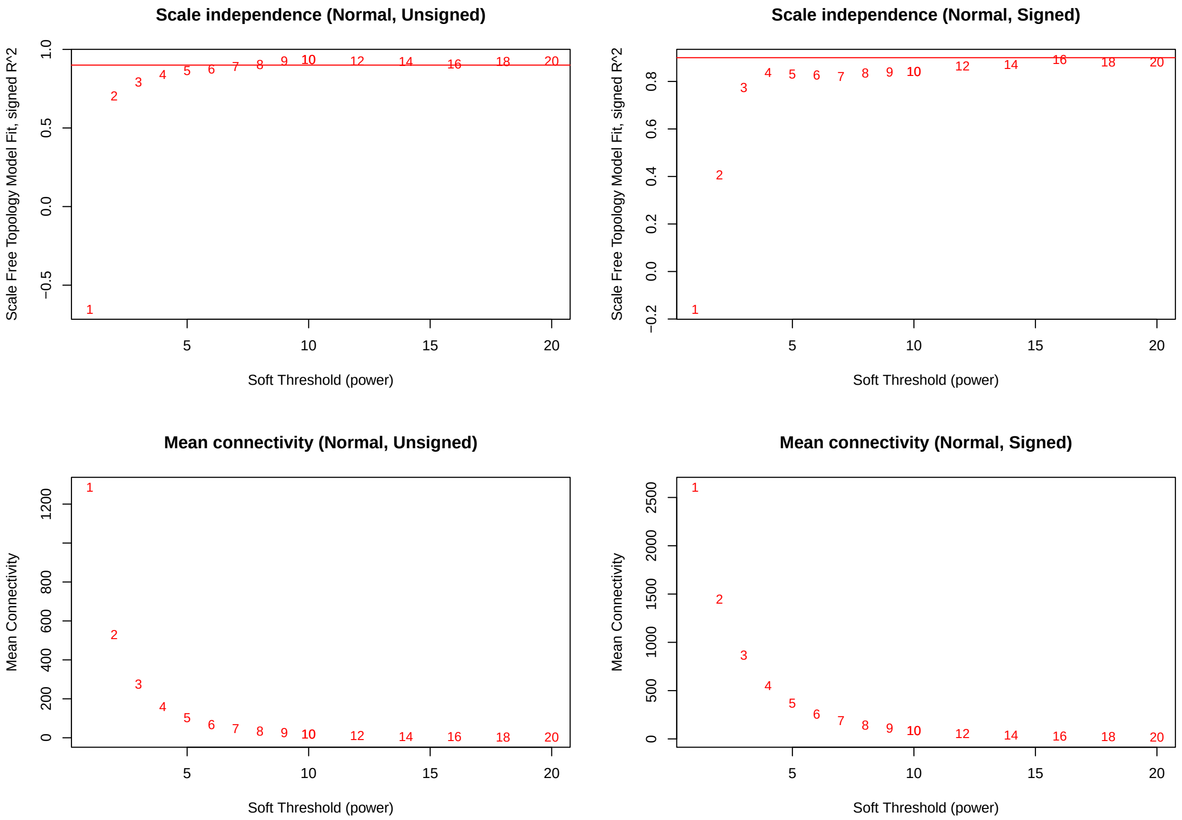


Figure 2: Scale independence and mean connectivity plots for the **Normal** dataset. Both unsigned and signed configurations are shown. The higher powers required to cross the 0.90 threshold reflect the quieter, more homeostatic co-expression landscape of healthy tissue.

5 Co-expression Network Construction

5.1 How WGCNA Works

WGCNA transforms the raw expression matrix into a biological network through three sequential mathematical operations:

1. Pearson Correlation Matrix: For every pair of the x number of filtered genes, the Pearson correlation coefficient r_{ij} is calculated across all samples. This yields a $x \times x$ symmetric matrix with values between -1 and $+1$.

2. Adjacency Matrix: The correlation matrix is transformed by raising each element to the soft-thresholding power β. This suppresses weak correlations toward zero and retains strong ones. In the signed formulation: $A_{ij} = \left(\frac{1+r_{ij}}{2}\right)^{\beta}$. In the unsigned formulation: $A_{ij} = |r_{ij}|^{\beta}$.

3. Topological Overlap Matrix (TOM): The adjacency matrix is further refined by computing Topological Overlap, a measure that upgrades the connection strength between two genes if they share many common interaction partners. Formally: $\text{TOM}_{ij} = \frac{\sum_u A_{iu}A_{uj}+A_{ij}}{\min(k_i,k_j)+1-A_{ij}}$, where $k_i = \sum_u A_{iu}$ is the connectivity of gene i.

TOM is more robust than raw adjacency because it reflects the broader network context of each gene pair, not merely their pairwise correlation.

5.2 Module Detection with blockwiseModules and Tuning

The `blockwiseModules()` function executes the entire pipeline from raw expression matrix to final colour-coded modules. Key parameters:

```
netwk_tumor_signed <- blockwiseModules(
  t(filtered_tumor), # Transpose: samples as rows, genes as columns
  power = candidate_power, # Validated soft-thresholding power you are
      testing
  networkType = "signed",
  deepSplit = 2, # Moderate sensitivity for tree cutting
  minModuleSize = 30, # Minimum 30 genes per module
  TOMType = "signed",
  mergeCutHeight = 0.25, # Merge modules with eigengene correlation > 0.75
  numericLabels = TRUE, # Assign numeric IDs during computation
  saveTOMs = TRUE,
  saveTOMFileBase= "tumor_signed",
  verbose = 3
)
```

We construct networks for all our candidate powers (**total 6 networks constructed here initially, and 4 extra tuned ones**).

In the first run of the blockwiseModules function, exact hyperparameters mentioned in the snippet above were used. However, this run was found to give a very large number of modules, and also some in the end with less number of genes, which is generally not preferred. We want somewhere between 5-12 modules with a sort of balanced distribution among the modules. 'Balanced' here means that there should not be a huge disparity in the number of genes present in each module, the range of difference should be reasonable.

The above output was still helpful for us in terms of finalising which of the candidate power to use for the normal signed and unsigned networks. The finalised powers have been written in bold text in the candidate powers table.

To get over this issue, we performed hyperparameter tuning. **The value of deepSplit was changed from 2 to 1 and minModuleSize was changed from 30 to 50**. The final module sizes of the tuned networks have been given in the table below.

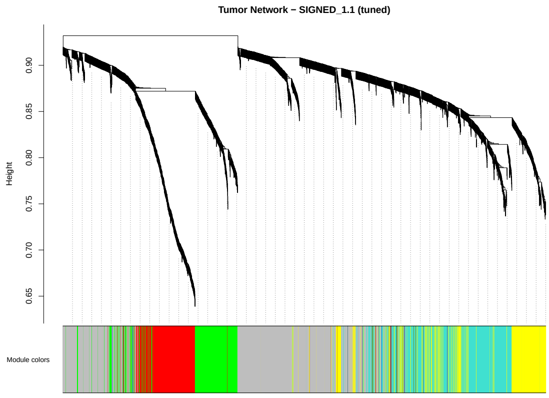
5.3 Module Size Summary (Tuned and Finalised Networks)

After running `blockwiseModules()` for all four configurations, the following module structures were obtained:

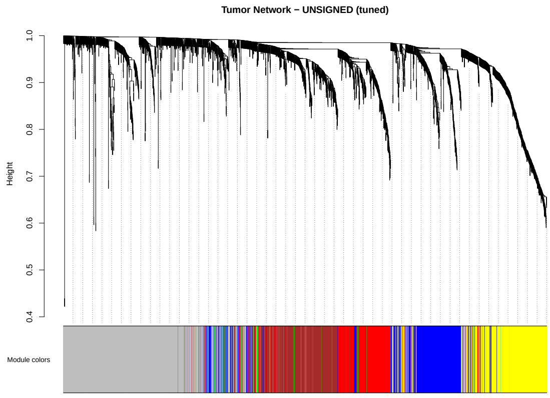
Configuration	Grey	Module 1	Module 2	Module 3	Module 4	Module 5	Module 6	Module 7	Module 8
Tumor Unsigned	1689	838	511	491	475	446	349	288	70
Tumor Signed	1798	650	609	590	463	385	360	302	–
Normal Unsigned	1276	2263	762	297	251	175	133	–	–
Normal Signed	1190	1656	1269	358	258	257	169	–	–

The **grey module** (Module 0) contains genes that could not be assigned to any functional co-expression group. It is treated as background noise and excluded from all downstream biological analyses.

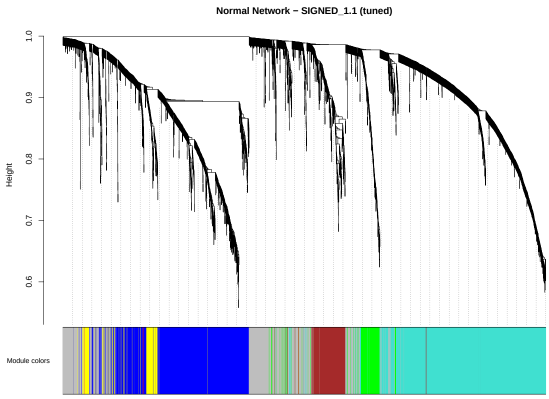
6 Figure 2: Cluster Dendrograms



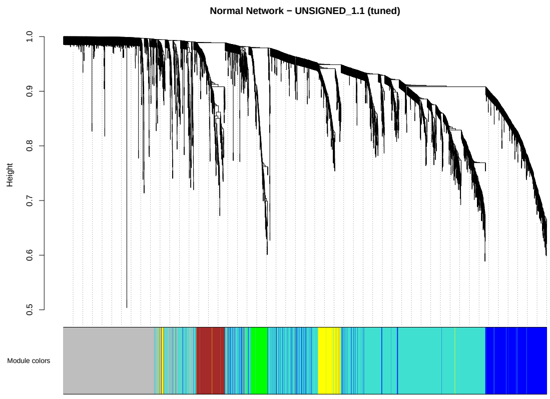
(a) Tumor Network – SIGNED TUNED



(b) Tumor Network – UNSIGNED TUNED



(c) Normal Network – SIGNED TUNED



(d) Normal Network – UNSIGNED TUNED

Figure 3: Cluster dendrograms for all 4 final, tuned network configurations.

6.1 Anatomy of a Dendrogram

Each dendrogram is produced by applying average-linkage hierarchical clustering to the TOM dissimilarity matrix ($1-\mathrm{TOM}$). Every vertical line at the bottom of the tree represents one of the genes. Genes merge into twigs, twigs into branches, and branches into a central trunk. The height at which two branches merge reflects their topological distance: a merger at height 0.3 means the two gene groups share many common interaction partners and behave very similarly; a merger near height 1.0 indicates essentially independent expression programmes.

The colour bar beneath the tree shows the module colour assigned to each gene. Genes within the same colour block form a co-expression module.

How are they constructed?

1. Hierarchial Clustering: The algorithm treats every gene as an individual leaf. It looks at the generated TOM and begins branching highly similar genes together into small twigs. These twigs merge into branches and ranches merge into a central tree trunk.

2. Dynamic Tree Cutting: WGCNA runs a sliding blade across these branches governed by the deepSplit and minModuleSize settings we saw before. If a branch is distinct and has enough leaves, it gets sliced off and assigned its own unique module colour.

7 Module Eigengenes and Correlation Heatmaps

7.1 What Is a Module Eigengene and Why Do We Need It?

After network construction, each module may contain hundreds of genes. Tracking every individual gene's expression across all samples simultaneously is both computationally unwieldy and visually impossible. WGCNA solves this by reducing each module to a single representative expression vector called the **Module Eigengene (ME)**.

Mathematical definition: The Module Eigengene of a given module is defined as the **first principal component** of the gene expression matrix formed by all genes in that module. Principal Component Analysis (PCA) finds the linear combination of variables (genes) that explains the maximum possible variance in the data. The first principal component is therefore the single axis in expression space that best summarises the overall activity trend of the module across all samples.

A high ME value for a sample means most genes in the module are relatively highly expressed in that sample; a low value means they are suppressed.

This is why the ME is often described as the "representative expression profile" or the "ambassador" of the module: it faithfully summarises the collective behaviour of all module members in a single trackable quantity.

7.2 Inter-Module Eigengene Correlations

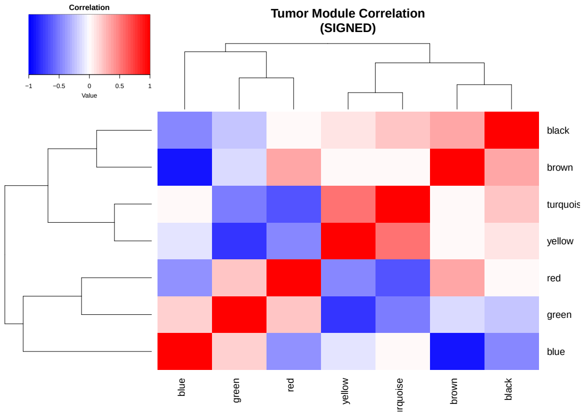
Once an ME is computed for every module, the pairwise Pearson correlation coefficients between all ME pairs are calculated. This produces a small, square correlation matrix (one row and column per module). The interpretation is straightforward:

- A **strong positive correlation** between two module eigengenes (cell colour red, $r \approx +1$) means the two pathways tend to be co-activated across samples. When one module's genes are highly expressed, the other module's genes are too. This suggests the two modules participate in a shared or co-regulated higher-level biological programme.
- A **strong negative correlation** (cell colour blue, $r \approx -1$) means the two pathways are antagonistic: one is consistently suppressed when the other is active. This is the mathematical fingerprint of a biological inhibitory or feedback relationship.
- A **near-zero correlation** (cell colour white) means the two modules operate essentially independently.

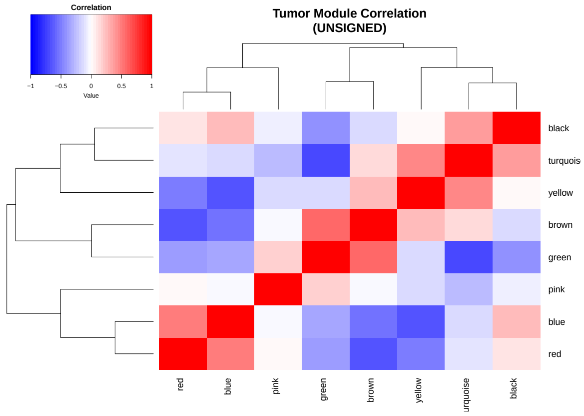
7.3 Grey Module Exclusion

The grey module (Module 0) contains all genes that could not be confidently assigned to any co-expression group. Including it in the eigengene correlation analysis would contaminate the heatmap with a biologically uninformative "noise" vector. The grey column is therefore explicitly removed from the eigengene matrix before correlation computation and heatmap generation:

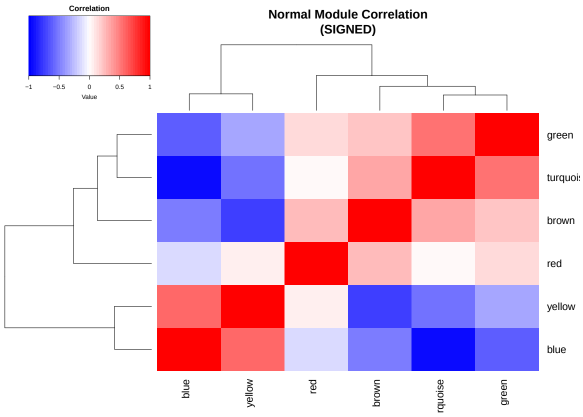
8 Figure 3: Module Eigengene Correlation Heatmaps



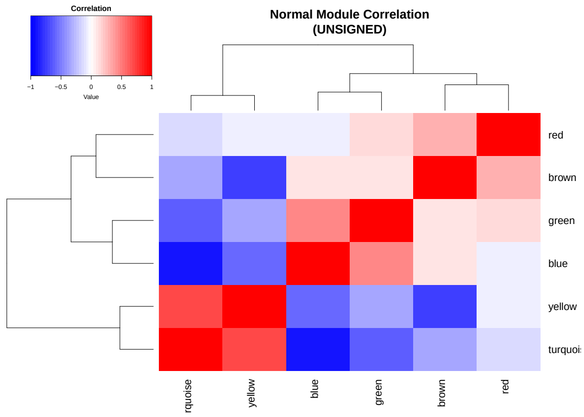
(a) Tumor – SIGNED



(b) Tumor – UNSIGNED



(c) Normal – SIGNED



(d) Normal – UNSIGNED

Figure 4: Pairwise Pearson correlation heatmaps of module eigengenes for all four finalised network configurations. Red cells indicate strong positive eigengene correlation; blue cells indicate anti-correlation. The colour scale runs from -1 (deep blue) through 0 (white) to $+1$ (deep red). The dendrograms on the axes reflect hierarchical clustering of the modules by eigengene similarity, so similar modules appear adjacent.

9 Module Preservation Analysis

9.1 Biological Motivation

At this stage the analysis has produced separate co-expression networks for tumour and normal tissue. Now we want to start to visualise some sort of connection/relation between the tumor and normal conditions.

A tumour module that retains its co-expression structure in normal tissue is not a cancer-specific phenomenon; it likely represents a fundamental cellular pathway that is active regardless of disease state. By contrast, **a tumour module whose internal organisation collapses in normal tissue is genuinely disease-specific**. These disease-specific modules are the most biologically interesting candidates for biomarker identification and therapeutic targeting.

9.2 The Permutation-Based Z-summary Statistic

How modulePreservation() works:

The function takes the tumour network as the **reference** (the network whose modules are being tested) and the normal network as the **test** dataset (the context in which preservation is evaluated). For each tumour module, it calculates a set of network topology statistics, including:

- **Module density:** how tightly the module's genes are interconnected with each other in the test dataset.
- **Module connectivity:** whether the genes that are highly connected hubs in the reference module are also well-connected in the test dataset.
- **Separability:** whether the module's genes are more similar to each other than to genes outside the module in the test dataset.

These observed statistics are then compared against a **null distribution** built by randomly permuting gene labels 100 times. Each permutation scrambles which genes belong to which "module" while keeping the expression data fixed. By doing this 100 times, the function builds an empirical picture of what these topology statistics look like by chance alone, if module assignments were random.

The **Z-score** for each statistic is then:

$$Z = \frac{\text{observed statistic} - \text{mean of permuted statistics}}{\text{standard deviation of permuted statistics}}$$

This measures how many standard deviations above the random baseline the real module sits. A large positive Z-score means the real module's topology statistics are far above what random chance would predict, so the module is genuinely preserved.

The **Z-summary** is a composite score that aggregates multiple individual Z-scores (density, connectivity, separability) into one summary number. It is the primary statistic used to judge preservation.

9.3 Interpretation Thresholds

The Z-summary score is interpreted using the following empirically established thresholds:

Z-summary	Category	Interpretation
< 2	No preservation	The module's co-expression structure collapses completely in normal tissue. The genes are not co-expressed in healthy cells. This is a tumour-specific module, likely driven by oncogenic mutation or epigenetic reprogramming.
$2 - 10$	Moderate preservation	The core co-expression relationships partially survive in normal tissue but the full module architecture is significantly altered. These modules represent pathways that exist in both contexts but have been substantially rewired in cancer.
> 10	Strong preservation	The module's topology is essentially intact in normal tissue. These modules represent fundamental cellular pathways (housekeeping functions, basic metabolism, structural maintenance) that are active regardless of disease state and are therefore less useful as cancer-specific biomarkers.

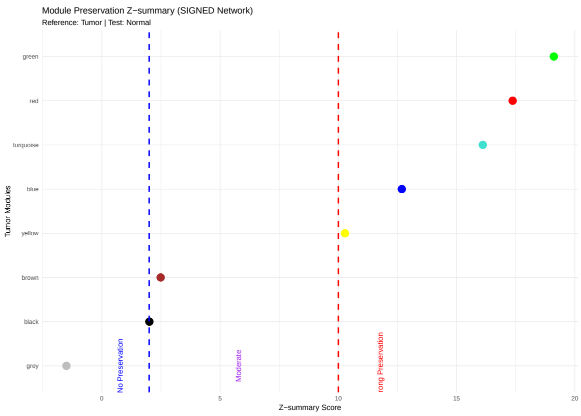
9.4 Data Quality Check Before Preservation Analysis

Before running `modulePreservation()`, the function `goodSamplesGenesMS()` is called to identify any genes with zero variance or excessive missing values across the multi-set data structure. Even after VST and filtering, it is possible for one gene to have exactly zero variance across one of the two cohorts (e.g. a gene that happened to be uniformly expressed across all 113 normal samples after transformation). Such a gene cannot contribute to any correlation calculation and would cause the preservation function to crash. The check identified and removed exactly **1 such gene** from the multi-set structure before preservation was run:

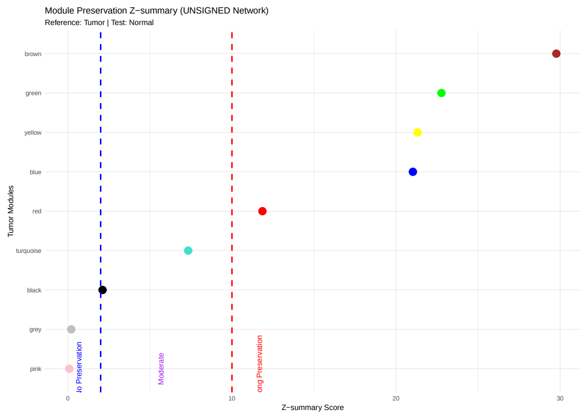
```
gsg <- goodSamplesGenesMS(multiData_signed, verbose = 3)
# Output: Excluding 1 gene due to zero variance

if (!gsg$allOK) {
  multiData_signed$Tumor$data <-
    multiData_signed$Tumor$data[gsg$goodSamples[[1]], gsg$goodGenes]
  multiData_signed$Normal$data <-
    multiData_signed$Normal$data[gsg$goodSamples[[2]], gsg$goodGenes]
  remaining_genes <- colnames(multiData_signed$Tumor$data)
  multiColor_signed$Tumor <- multiColor_signed$Tumor[remaining_genes]
  multiColor_signed$Normal <- multiColor_signed$Normal[remaining_genes]
}
```

10 Figure 4: Module Preservation Z-summary Plots



(a) Signed Network



(b) Unsigned Network

Figure 5: Module preservation Z-summary dot plots for signed (left) and unsigned (right) tumour networks, tested against the normal tissue dataset. Each coloured dot represents one tumour module, with the dot colour matching the WGCNA module colour. Modules are ordered from lowest (top) to highest (bottom) Z-summary score. The left dashed blue line marks $Z = 2$ (boundary between no preservation and moderate preservation). The right dashed red line marks $Z = 10$ (boundary between moderate and strong preservation). The grey module is included in the plot for completeness but is excluded from all biological interpretation.

10.1 Signed Network Preservation Results

The most biologically interesting finding from the signed analysis is the relative vulnerability of the **black** and **brown** modules. Their low Z-summary scores indicate that the specific co-expression relationships that define these modules in the tumour context do not hold together in healthy breast tissue. These are therefore candidates for tumour-specific pathway analysis.

10.2 Unsigned Network Preservation Results

The unsigned Z-summary plot (Figure 5b) reveals broadly similar patterns but with some important differences:

No preservation:

- **Pink** ($Z \approx 0.09$): This small module (70 genes), which only appeared under unsigned network rules, shows essentially zero preservation in normal tissue. Its genes are co-expressed only in the tumour context when correlation direction is ignored. This is consistent with the expectation that unsigned-specific modules are often driven by mixed positive and negative correlations that do not reflect coherent biological co-regulation.

Strong preservation:

- **Brown** ($Z \approx 29.8$): Exceptionally strongly preserved in the unsigned network, in stark contrast to its moderate preservation in the signed network. This discrepancy is informative: it suggests that the brown module contains genes that are co-expressed in both tumour and normal tissue, but that a subset of those co-expression relationships are in opposite directions between the two conditions. The unsigned network lumps these together and sees strong preservation; the signed network separates them and correctly identifies that the directional structure is less conserved.

- **Blue** ($Z \approx 21.0$), **Green** ($Z \approx 22.8$), **Yellow** ($Z \approx 22.8$): All strongly preserved.

The **black module's** low preservation score is consistent across both signed and unsigned analyses, reinforcing its status as the most tumour-specific module in this dataset and the highest priority for downstream biological investigation.

Please find the remaining part (Gene Ontology Enrichment analysis and Hub Gene Identification) in the next report, i.e., BRCA_WGCNA_Report_2.pdf.